Tan Yuan Xiang Elroy

elroytanyuanxiang@gmail.com | 8838 5573 | Singapore | tanyuanxiangelroy.github.io | elroytan

Summary

I'm Elroy, a 23-year-old guy on a journey of self-discovery through the exciting worlds of IT and engineering. Currently calling Singapore home, I'm taking a detour from the typical academic path to explore and develop myself before diving into SUTD for a degree.

Experience

ST Engineering, Cybersecurity Software Development & Quality Assurance Internship — Singapore
Aug 2026 – Jan 2027 · 6 months

- An on-site cybersecurity software development and quality assurance internship in Singapore.

BasedApp, Internship — Singapore
Sept 2025 – Mar 2026 · 7 months

- Engineered a TypeScript MCP (Model Context Protocol) integration with the Hyperliquid API to query public databases and validate on-chain transactions, cutting manual debugging time for user inquiries.
- Led the migration to Intercom, designing automated workflows and AI-triage agents supporting 12,000+ monthly active users and personally resolving 470+ technical incidents a month.
- Managed Discord community security (grew 4,000 to 6,000+ members): deployed anti-phishing bots, audited role permissions, and ran scam incident response.
- Authored the "Community Response Playbook" and maintained the official GitBook docs as the single source of truth for the support team.
- Built an internal Telegram + Hyperliquid MCP server (Node.js, TypeScript) with chat allow-listing and secure session management, enabling AI agents to query on-chain positions/order history within API rate limits (1,200 weight/min).

Singapore Army, Supply Assistant (SA) — Singapore
July 2023 – May 2025 · 1 year 11 months

- Managed TALC (TRADOC Admin and Logistic Center) logistic stores: procurement and maintaining essential stock levels and supplies.
- Volunteered 3+ weeks on attachment to ATEC for Exercise Wallaby (XWB) 2024; awarded the ATEC commander's coin for outstanding service.

Singapore Polytechnic, IoT Engineering Internship — Singapore
Feb 2022 – Aug 2022 · 7 months

- Led a team of 3 to build an automated mushroom-farming system, integrating hardware sensors with software monitoring.
- Owned the hardware design and integrated the software into the system.

Education

Bachelor of Engineering (Computer Science and Design) — **SUTD**, Computer and Information Systems Security / Information Assurance
May 2025 – Apr 2028

- GPA: 4.12/5.0
- Awarded the SUTD Global Merit Scholarship
- Member of SOAR (planning team of SOAR Challenge 2025)
- Member of athletics/running club

Degree **SUTD**, SP-SUTD Pathway Programme — Sept 2022 – July 2023

- Active member in student clubs SOAR and MRS
- 2023 IC Outreach Line Tracing Workshop (Main Facilitator)
- Learn more about the SP-SUTD Pathway: SP-SUTD-NUS

Polytechnic Diploma **Singapore Polytechnic**, Computer Engineering — Dec 2020 – Dec 2023

- GPA: 3.91
- Selected for the SP-SUTD Pathway Programme (1 of only 30 students)

High School Diploma **Hua Yi Secondary School** — Dec 2015 – Dec 2019

Awards

- Diploma with Merit, Singapore Polytechnic (2023-05): Top 10% of each graduating course cohort.

Certifications

- CPR (Hands-Only) + AED Provider, Singapore Red Cross (2021-06 to 2023-06).
- Standard First Aid Provider, Singapore Red Cross (2021-06 to 2023-06).
- Microsoft Certified: Azure Fundamentals, Microsoft (2021-06).
- Docker Fundamentals, Dell Technologies (2026-09).
- Google Cybersecurity, Coursera (2024-05).
- CS50 Cybersecurity, CS50 (Harvard) (2024-03).

Skills

- Computer Hardware & Software Engineering
- Object-Oriented Programming (OOP)
- English (Native speaker)
- C++
- Java
- Python (Programming Language)
- Project Management
- Cybersecurity
- Docker
- Network Security